Complex Numbers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic with complex numbers**

Let $z_1 = 4 - 2i$ and $z_2 = -3 + i$. Find $z_1 + z_2$ and $z_1 z_2$.

$z_1 + z_2 = (4 - 3) + (-2 + 1)i = 1 - i$. $z_1 z_2 = (4)(-3) - (-2)(1) + [(4)(1) + (-2)(-3)]i = -10 + 10i$.

2 **[Foundational]** • **Target: Arithmetic with complex numbers**

Find $(6 + i) - (2 - 5i)$.

$= 4 + 6i$

3 **[Foundational]** • **Target: Complex roots of a quadratic**

Solve $x^2 + 6x + 13 = 0$.

$\Delta = 6^2 - 4(1)(13) = 36 - 52 = -16 < 0$. By Theorem 1.1, $x = \frac{-6 \pm \sqrt{-16}}{2} = \frac{-6 \pm 4i}{2} = -3 \pm 2i$.

4 **[Foundational]** • **Target: Complex roots of a quadratic**

Solve $2x^2 - 4x + 10 = 0$.

$\Delta = (-4)^2 - 4(2)(10) = 16 - 80 = -64 < 0$. $x = \frac{4 \pm \sqrt{-64}}{4} = \frac{4 \pm 8i}{4} = 1 \pm 2i$.

5 **[Foundational]** • **Target: Complex roots of a quadratic**

Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.

Its discriminant is $\Delta = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0$. A negative discriminant means the quadratic formula takes the square root of a negative number, giving a complex conjugate pair (Theorem 1.1) instead of real roots.

6 **[Foundational]** • **Target: Complex roots of a quadratic**

Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.

$(-3 + 2i)^2 = 9 - 12i + 4i^2 = 9 - 12i - 4 = 5 - 12i$. And $6(-3 + 2i) = -18 + 12i$. Adding: $(5 - 12i) + (-18 + 12i) + 13 = 0$. Yes, it checks out.

7 **[Foundational]** • **Target: Modulus**

Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).

$|z| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Modulus**

Find $|z|$ for $z = 5 - 12i$.

$|z| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

9 **[Intermediate]** • **Target: Modulus**

Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.

$|a + bi| = \sqrt{a^2 + b^2}$ and $|a - bi| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2}$ — squaring b or $-b$ gives the same result, so the two moduli are identical.

10 **[Intermediate]** • **Target: Modulus**

Let z_1 and z_2 be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number w such that $w = z_1^2 + z_2^2$. If w is expressed in the form $a + bi$, where a and b are real numbers, find the value of $a - b$.

The quadratic equation is $x^2 - 4x + 5 = 0$. By Vieta's formulas, the sum of the roots is $z_1 + z_2 = -(-4)/1 = 4$ and the product of the roots is $z_1 z_2 = 5/1 = 5$. We are interested in $w = z_1^2 + z_2^2$. We can rewrite this as $w = (z_1 + z_2)^2 - 2z_1 z_2$. Substituting the values from Vieta's formulas: $w = (4)^2 - 2(5)$ $w = 16 - 10$ $w = 6$. So, $w = 6$. When expressed in the form $a + bi$, we have $a = 6$ and $b = 0$. The question asks for $a - b$. $a - b = 6 - 0 = 6$.

11 **[Intermediate]** • **Target: Modulus-Argument Form**

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.

Let $w = (a + bi) + (3 - 2i) = (a + 3) + (b - 2)i$. We are given that $|w|^2 = 25$, so $(a + 3)^2 + (b - 2)^2 = 25$. Since w lies in the first quadrant, $a + 3 > 0$ and $b - 2 > 0$. We need to find integer or simple fractional solutions for $(a + 3)$ and $(b - 2)$ that sum of squares is 25. The possible pairs of squares are $(9, 16)$ and $(16, 9)$. \\ Case 1: $(a + 3)^2 = 9$ and $(b - 2)^2 = 16$. This gives $a + 3 = \pm 3$ and $b - 2 = \pm 4$. Since $a + 3 > 0$ and $b - 2 > 0$, we have $a + 3 = 3$ and $b - 2 = 4$. This leads to $a = 0$ and $b = 6$. However, if $a = 0$, then $w = 3 + 4i$, and $a + b = 0 + 6 = 6$. \\ Case 2: $(a + 3)^2 = 16$ and $(b - 2)^2 = 9$. This gives $a + 3 = \pm 4$ and $b - 2 = \pm 3$. Since $a + 3 > 0$ and $b - 2 > 0$, we have $a + 3 = 4$ and $b - 2 = 3$. This leads to $a = 1$ and $b = 5$. In this case, $w = 4 + 3i$, which is in the first quadrant. We want to find $a + b$. So, $a + b = 1 + 5 = 6$. \\ Both cases yield $a + b = 6$.

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12 [Intermediate] • Target: De Moivre's Powers

Let $z_1 = 2 + 3i$ and $z_2 = 1 - i$. Consider the complex number w such that $w = \frac{z_1^2}{z_2}$. If $w = a + bi$, where a and b are real numbers, find the value of b.

First, calculate z_1^2: $z_1^2 = (2 + 3i)^2 = 2^2 + 2(2)(3i) + (3i)^2 = 4 + 12i + 9i^2 = 4 + 12i - 9 = -5 + 12i$. Next, divide z_1^2 by z_2: $w = \frac{-5 + 12i}{1 - i}$. To simplify this division, multiply the numerator and the denominator by the conjugate of the denominator, which is $1 + i$: $w = \frac{-5 + 12i}{1 - i} \times \frac{1 + i}{1 + i} = \frac{(-5 + 12i)(1 + i)}{(1 - i)(1 + i)}$. The numerator is: $(-5)(1) + (-5)(i) + (12i)(1) + (12i)(i) = -5 - 5i + 12i + 12i^2 = -5 + 7i - 12 = -17 + 7i$. The denominator is: $(1)^2 - (i)^2 = 1 - (-1) = 1 + 1 = 2$. So, $w = \frac{-17 + 7i}{2} = -\frac{17}{2} + \frac{7}{2}i$. Given that $w = a + bi$, we have $a = -\frac{17}{2}$ and $b = \frac{7}{2}$. The question asks for the value of b. Therefore, $b = \frac{7}{2}$. Final Answer: $-\frac{17}{2}$

13 [Intermediate] • Target: Roots of Complex Numbers

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

14 [Intermediate] • Target: Complex Geometry & Loci

Consider the function $g(x) = ax^2 + bx + c$. The graph of this quadratic function is shown below. The vertex of the parabola is at $(-2, 4)$. The parabola passes through the point $(0, -8)$. Find the value of the parameter a.

The general form of a quadratic function is $g(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $g(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(-2, 4)$, we can write the function as $g(x) = a(x - (-2))^2 + 4$, which simplifies to $g(x) = a(x + 2)^2 + 4$. We are also given that the parabola passes through the point $(0, -8)$. We can substitute these coordinates into the vertex form of the equation to solve for a. $-8 = a(0 + 2)^2 + 4$ $-8 = a(2)^2 + 4$ $-8 = 4a + 4$ Subtract 4 from both sides: $-8 - 4 = 4a$ $-12 = 4a$ Divide by 4: $a = \frac{-12}{4}$ $a = -3$ *Common Trap:* A common mistake is to assume the parabola passes through the origin or to confuse the vertex coordinates with other points. Another mistake is to incorrectly expand the squared term or solve the linear equation for 'a'. For example, if one were to incorrectly use $a = a(x - h)^2 + k$ and substitute $x = 0$ and $y = -8$ into the expanded form of the general quadratic without first establishing the relationship between the general coefficients and the vertex form, they might arrive at an incorrect value for a. For instance, if one incorrectly assumes $c = -8$ and then tries to relate b to the vertex's x-coordinate using $-b/(2a) = -2$ without considering the vertex's y-coordinate, they might get stuck or derive an incorrect 'a'.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Arithmetic with complex numbers • Cross-Lesson: 6.01 Probability

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. In this case, $a = 1$ and the x-coordinate of the vertex is given as 1. So, $1 = -\frac{b}{2(1)}$. This simplifies to $1 = -\frac{b}{2}$, which means $b = -2$. Alternatively, we can use the vertex form of a quadratic equation: $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Here, $a = 1$ and $(h, k) = (1, -4)$, so $f(x) = 1(x - 1)^2 - 4$. Expanding this, we get $f(x) = (x^2 - 2x + 1) - 4 = x^2 - 2x - 3$. Comparing this to $f(x) = x^2 + bx + c$, we see that $b = -2$ and $c = -3$. We can verify that the point $(-2, 5)$ lies on this parabola: $f(-2) = (-2)^2 - 2(-2) - 3 = 4 + 4 - 3 = 5$. This confirms our values. Common errors might involve misinterpreting the vertex formula or making sign errors during expansion.

16 [Challenging] • Target: Complex roots of a quadratic • Cross-Lesson: 2.01 Sequences & Series

Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?

The common difference of an arithmetic sequence is the constant value added to each term to get the next term. Observing the number line, we can see the sequence starts at 0 and progresses to 3, then to 6, and finally to 12. The difference between consecutive terms is $3 - 0 = 3$, $6 - 3 = 3$, and $12 - 6 = 6$. However, the diagram shows markings at 0, 3, 6, and 9. Let's re-examine the number line to ensure accuracy. The marks are at 0, 3, 6, and 9. The difference between consecutive terms is $3 - 0 = 3$, $6 - 3 = 3$, and $9 - 6 = 3$. Therefore, the common difference is 3.

17 [Challenging] • Target: Modulus • Cross-Lesson: 3.01 Functions & Graphs

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, the function can be written as $f(x) = a(x - 2)^2 - 5$. We are also given that the y-intercept is $(0, -1)$. This means that when $x = 0$, $f(x) = -1$. Substituting these values into the vertex form: $-1 = a(0 - 2)^2 - 5$. This simplifies to $-1 = a(-2)^2 - 5$, which is $-1 = 4a - 5$. Adding 5 to both sides gives $4 = 4a$. Dividing by 4 yields $a = 1$.

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18 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 4.01 Matrices & Vectors

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n - 1)d$. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Plugging these values into the formula, we get: $a_4 = 5 + (4 - 1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the 4th term of the sequence is 14.

19 [Challenging] • Target: De Moivre's Powers • Cross-Lesson: 5.01 Introductory Calculus

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

The formula for the nth term of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. Given $a_1 = 7$ and $d = 5$, we want to find a_3. Substitute $n = 3$ into the formula: $a_3 = 7 + (3 - 1) \times 5$ $a_3 = 7 + (2) \times 5$ $a_3 = 7 + 10$ $a_3 = 17$. Therefore, the 3rd term is 17.

20 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 6.01 Probability

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

The magnitude of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the magnitude is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic with complex numbers

Let $z_1 = -6 + 9i$ and $z_2 = 2 + 6i$. Find $z_1 + z_2$.

a. $-2 + 15i$

b. $-4 + 13i$

c. $-4 + 15i$ ✓

d. $-8 + 3i$

$z_1 + z_2 = (-6 + 2) + (9 + 6)i = -4 + 15i$.

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22 [Foundational] • Target: Complex roots of a quadratic

Let $z_1 = 3 + 8i$ and $z_2 = -9 - 7i$. Find $z_1 + z_2$.

a. $12 + 15i$

b. $-6 - i$

c. $-4 + i$

d. $-6 + i$ ✓

$z_1 + z_2 = (3 - 9) + (8 - 7)i = -6 + i$.

23 [Foundational] • Target: Modulus

Let $z_1 = -8 - 8i$ and $z_2 = 5 + 3i$. Find $z_1 + z_2$.

a. $-13 - 11i$

b. $-3 - 5i$ ✓

c. $-1 - 5i$

d. $-3 - 7i$

$z_1 + z_2 = (-8 + 5) + (-8 + 3)i = -3 - 5i$.

24 [Foundational] • Target: Modulus-Argument Form

Let $z_1 = -7 - 3i$ and $z_2 = -4 + 2i$. Find $z_1 + z_2$.

a. $-3 - 5i$

b. $-11 - 3i$

c. $-9 - i$

d. $-11 - i$ ✓

$z_1 + z_2 = (-7 - 4) + (-3 + 2)i = -11 - i$.

25 [Foundational] • Target: De Moivre's Powers

Let $z_1 = 5 - i$ and $z_2 = -3 - 6i$. Find $z_1 + z_2$.

a. $2 - 9i$

b. $8 + 5i$

c. $2 - 7i$ ✓

d. $4 - 7i$

$z_1 + z_2 = (5 - 3) + (-1 - 6)i = 2 - 7i$.

26 [Foundational] • Target: Roots of Complex Numbers

Let $z_1 = 5 + 2i$ and $z_2 = 8 - 2i$. Find $z_1 + z_2$.

a. 13 ✓

b. $-3 + 4i$

c. $13 - 2i$

d. 15

$z_1 + z_2 = (5 + 8) + (2 - 2)i = 13$.

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27 [Foundational] • Target: Complex Geometry & Loci

Let $z_1 = -8 - i$ and $z_2 = -5 + 6i$. Find $z_1 + z_2$.

a. $-13 + 3i$

b. $-3 - 7i$

c. $-11 + 5i$

d. $-13 + 5i$ ✓

$z_1 + z_2 = (-8 - 5) + (-1 + 6)i = -13 + 5i$.

28 [Foundational] • Target: Arithmetic with complex numbers

Solve $x^2 - 2x + 5 = 0$.

a. $1 \pm 2i$ ✓

b. $1 \pm 3i$

c. $2 \pm 2i$

d. $-1 \pm 2i$

$\Delta = (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$. $x = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Complex roots of a quadratic

Solve $x^2 + 12x + 72 = 0$.

a. $6 \pm 6i$

b. $-6 \pm 6i$ ✓

c. $-6 \pm 7i$

d. $-5 \pm 6i$

$\Delta = 12^2 - 4(1)(72) = 144 - 288 = -144 < 0$. $x = \frac{-12 \pm \sqrt{-144}}{2} = \frac{-12 \pm 12i}{2} = -6 \pm 6i$.

30 [Intermediate] • Target: Modulus

Solve $x^2 - 6x + 58 = 0$.

a. $3 \pm 7i$ ✓

b. $4 \pm 7i$

c. $-3 \pm 7i$

d. $3 \pm 8i$

$\Delta = (-6)^2 - 4(1)(58) = 36 - 232 = -196 < 0$. $x = \frac{6 \pm \sqrt{-196}}{2} = \frac{6 \pm 14i}{2} = 3 \pm 7i$.

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31 [Intermediate] • Target: Modulus-Argument Form

Solve $x^2 - 10x + 41 = 0$.

a. $5 \pm 4i$ ✓

b. $-5 \pm 4i$

c. $5 \pm 5i$

d. $6 \pm 4i$

$\Delta = (-10)^2 - 4(1)(41) = 100 - 164 = -64 < 0$. $x = \frac{10 \pm \sqrt{-64}}{2} = \frac{10 \pm 8i}{2} = 5 \pm 4i$.

32 [Intermediate] • Target: De Moivre's Powers

Solve $x^2 - 6x + 45 = 0$.

a. $3 \pm 7i$

b. $4 \pm 6i$

c. $-3 \pm 6i$

d. $3 \pm 6i$ ✓

$\Delta = (-6)^2 - 4(1)(45) = 36 - 180 = -144 < 0$. $x = \frac{6 \pm \sqrt{-144}}{2} = \frac{6 \pm 12i}{2} = 3 \pm 6i$.

33 [Intermediate] • Target: Roots of Complex Numbers

Solve $x^2 - 16x + 113 = 0$.

a. $9 \pm 7i$

b. $8 \pm 8i$

c. $-8 \pm 7i$

d. $8 \pm 7i$ ✓

$\Delta = (-16)^2 - 4(1)(113) = 256 - 452 = -196 < 0$. $x = \frac{16 \pm \sqrt{-196}}{2} = \frac{16 \pm 14i}{2} = 8 \pm 7i$.

34 [Intermediate] • Target: Complex Geometry & Loci

Solve $x^2 + 16x + 80 = 0$.

a. $8 \pm 4i$

b. $-7 \pm 4i$

c. $-8 \pm 4i$ ✓

d. $-8 \pm 5i$

$\Delta = 16^2 - 4(1)(80) = 256 - 320 = -64 < 0$. $x = \frac{-16 \pm \sqrt{-64}}{2} = \frac{-16 \pm 8i}{2} = -8 \pm 4i$.

35 [Intermediate] • Target: Arithmetic with complex numbers

Find $|z|$ for $z = -1 + 6i$.

a. 37

b. $\sqrt{39}$

c. $\sqrt{35}$

d. $\sqrt{37}$ ✓

$|z| = \sqrt{(-1)^2 + 6^2} = \sqrt{37}$.

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36 [Intermediate] • Target: Complex roots of a quadratic

Find $|z|$ for $z = -12 - 35i$.

a. 47

b. 39

c. 35

d. 37 ✓

$|z| = \sqrt{(-12)^2 + (-35)^2} = \sqrt{1369} = 37$.

37 [Intermediate] • Target: Modulus

Find $(2 - 7i) - (-6 - 9i) + (-8 - 5i)$.

a. $0 - 5i$

b. $3 - 3i$

c. $0 - 3i$ ✓

d. $2 - 3i$

$(2 + 6 - 8) + (-7 + 9 - 5)i = 0 - 3i$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 4.01 Matrices & Vectors

Find $(8 + i) - (5 - 3i) + (-8 + 9i)$.

a. $-5 + 11i$

b. $-5 + 13i$ ✓

c. $-3 + 13i$

d. $5 + 13i$

$(8 - 5 - 8) + (1 + 3 + 9)i = -5 + 13i$.

39 [Challenging] • Target: De Moivre's Powers • Cross-Lesson: 5.01 Introductory Calculus

Multiply $(3 - i)(6 + 7i)$.

a. $25 + 27i$

b. $25 + 15i$ ✓

c. $25 + 17i$

d. $11 + 15i$

$18 + 21i - 6i - 7i^2 = 18 + 15i - 7(-1) = 25 + 15i$.

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40 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 6.01 Probability

Multiply $(6-4i)(-9+i)$.

a. $-50+44i$

b. $-50-30i$

c. $-50+42i$ ✓

d. $-58+42i$

$-54+6i+36i-4i^2 = -54+42i-4(-1) = -50+42i$.

41 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 2.01 Sequences & Series

Find the polar representation of $z = -1-i\sqrt{3}$.

a. $2(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3})$ ✓

b. $2(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3})$

c. $2(\cos\frac{5\pi}{3}+i\sin\frac{5\pi}{3})$

d. $4(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3})$

$r=\sqrt{(-1)^2+(-\sqrt{3})^2}=\sqrt{4}=2$. Angle is in Quadrant III: $\theta=\pi+\frac{\pi}{3}=\frac{4\pi}{3}$.

42 [Challenging] • Target: Modulus-Argument Form • Cross-Lesson: 3.01 Functions & Graphs

If $z=4(\cos 70^\circ+i\sin 70^\circ)$ and $w=2(\cos 40^\circ+i\sin 40^\circ)$, what is $\frac{z}{w}$?

a. $2(\cos 30^\circ+i\sin 30^\circ)$ ✓

b. $2(\cos 110^\circ+i\sin 110^\circ)$

c. $8(\cos 30^\circ+i\sin 30^\circ)$

d. $2(\cos 2800^\circ+i\sin 2800^\circ)$

Divide moduli: $\frac{4}{2}=2$. Subtract angles: $70^\circ-40^\circ=30^\circ$.

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43 [Challenging] • Target: De Moivre's Powers • Cross-Lesson: 4.01 Matrices & Vectors

If $z + \frac{1}{z} = 2\cos\theta$, what is $z^n + \frac{1}{z^n}$?

a. $2\cos(n\theta)$ ✓

b. $2\sin(n\theta)$

c. $\cos(n\theta)$

d. $2^n\cos^n\theta$

Since $z = \cos\theta + i\sin\theta$, $z^n = \cos(n\theta) + i\sin(n\theta)$ and $z^{-n} = \cos(n\theta) - i\sin(n\theta)$. Adding gives $2\cos(n\theta)$.

44 [Challenging] • Target: Roots of Complex Numbers • Cross-Lesson: 5.01 Introductory Calculus

The four 4th roots of -1 form the vertices of what geometric polygon in the complex plane?

a. **A square** ✓

b. A rectangle

c. A rhombus

d. A regular hexagon

The n-th roots of any non-zero complex number are evenly spaced on a circle with angle separation $\frac{2\pi}{n} = \frac{\pi}{2}$, forming a square.

45 [Challenging] • Target: Complex Geometry & Loci • Cross-Lesson: 6.01 Probability

Describe the locus of points in the complex plane satisfying $|z - 3| = |z - 5i|$.

a. **Perpendicular bisector of segment joining** $(3, 0)$ **and** $(0, 5)$ ✓

b. Circle of radius 3

c. Ellipse with foci at 3 and $5i$

d. Ray starting at 3

The equation states that the distance from z to 3 equals the distance from z to $5i$, which is the perpendicular bisector.

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